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Pathways to commercially viable organic photovoltaic materials

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Organic photovoltaics (OPVs) are a pathway to sustainable energy solutions in various applications, but the challenge of developing materials that simultaneously fulfil stringent cost, efficiency and stability requirements has limited widespread adoption. Here we examine the critical factors shaping the transition of OPV materials from laboratory research to real-world deployment, focusing on materials design, scalable manufacturing and device reliability. Recent laboratory-scale proof-of-concept and prototype demonstrations have advanced the development of OPV materials with device efficiencies that exceed 21%, yet overcoming scale-up challenges remains essential for commercial viability. To facilitate this lab-to-fab transition, we discuss four key aspects that are expected to define the next decade of sustainable OPV: cost-effectiveness, green solvents for processing, stability and efficiency. By integrating these considerations, we highlight the advantages of OPV materials, including high power-to-weight ratios, intrinsic mechanical flexibility and tailored spectral selectivity for greenhouse agrivoltaics, to accelerate their commercialization.

Organic photovoltaics (OPVs) are a promising solution for sustainable energy, offering low-cost fabrication, lightweight design and mechanical flexibility. Over the past few decades, the power conversion efficiency (PCE) of OPVs has increased from below 1% to over 20% (refs. 1–5). Numerous strategies are being explored to advance OPV technology, leading to efficiencies approaching and exceeding 21% at the laboratory scale[2,3,5,6]. However, the transition from laboratory-scale breakthroughs to commercial viability remains a hurdle. The key questions are: can OPVs move beyond academic achievement to real-world deployment and which strategies will drive this transition?

Among the key components of OPVs, the photoactive layer (PAL; composed of organic semiconductor materials) is crucial for efficient photo-to-electric conversion while enabling high energy-to-weight ratios and mechanical flexibility. The life cycle of organic semiconductor materials used in PALs encompasses three interconnected stages: synthesis, processing for manufacturing and adoption in practical applications. Progress has been made in each of these areas, including the development of low-cost polymer donor and acceptor materials, green solvents for processing materials and the demonstration of various OPV applications[7–13]. Compared with silicon, CdTe and perovskite photovoltaics, OPVs offer advantages for agrivoltaics owing to their semi-transparency and PAL compositions that lack toxic metals. The tunable absorption spectra of organic semiconductor materials in the visible and near-infrared regions also facilitate selective spectral transmission, optimizing photosynthetically active radiation for the growth of various crops and positioning OPVs as an energy source for agricultural settings[9].

Some prototypes and demonstrations have been introduced to the public. These demonstrations include building-integrated photovoltaics installed in windows and hybrid plastic–glass OPV greenhouses[11,14–17]. Indoor OPVs for Internet of Things applications[18], self-powered ultra-flexible electronics based on nanograting-patterned OPVs for

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cardiac monitoring[7] and flexible OPV-integrated thermoregulatory textiles have also been realized at the prototype scale[12,19]. Furthermore, OPV applications have been explored on space platforms[20–22] and high-altitude stations[23,24], as well as in underwater environments[25,26]. However, despite three decades of research, the widespread commercial adoption of OPVs remains limited.

In this Review, we examine the challenges and opportunities associated with advancing OPV materials (referring to organic semiconductor materials in PALs) in scalable and commercially viable electronic devices. We emphasize the role of designing low-cost materials and using environmentally friendly green solvents for materials processing as strategies to bridge the lab-to-fab gap[27]. Key aspects of materials stability and efficiency are also discussed, and we focus on the primary factors that limit the large-scale commercialization of OPV technologies. We first explore four key aspects from a materials perspective: cost-effectiveness, green-solvent processability, stability and efficiency. Next, we highlight the advantages of OPV materials for real-world applications, which could complement other photovoltaic technologies for widespread adoption. Finally, we conclude with an outlook on future developments.

Materials synthesis and cost-effectiveness

Organic semiconductor materials enable low-cost production as they can be solution-processed via the high-speed and high-throughput capabilities of roll-to-roll manufacturing techniques[28–31]. Cost analyses of OPV manufacturing processes reveal that materials costs dominate the overall expenses[13,28,29,31]. Here we selected high-speed, layer-by-layer sequential deposition as the method of choice for evaluating the cost of state-of-the-art OPV materials[31] (Fig. 1a). By shifting module manufacturing from bulk heterojunction processing to the high-speed layer-by-layer sequential deposition technique, production costs of OPVs can be reduced by up to 75%, potentially bringing costs below those of silicon and perovskite photovoltaics for modules with comparably large areas (10^2–10^3 cm^2). Silicon is one of the most abundant elements on Earth and is valued for its intrinsic semiconductor properties, which have enabled widespread industrial applications. Its production costs have been continuously optimized over decades. In contrast, OPVs incorporating emerging non-fullerene acceptors (NFAs) are in their infancy. While a direct comparison of materials costs between silicon photovoltaics and OPVs is challenging, it is evident that reductions in these costs are essential for improving the commercial viability of OPV technology[13,29,31,32]. For a 7.5 cm^2 module (Fig. 1b), materials accounted for 76.3% of the total cost, with the primary contributors being PAL materials (36.4% for polymer donor PM6 and acceptor T8, the full names and chemical structures of which are provided in Supplementary Note 1 and Supplementary Fig. 1, respectively), indium tin oxide (ITO, 23.6%) and Ag electrodes (14.4%). The organic and perovskite photovoltaic fields have demonstrated that materials costs for electrodes can be reduced through ITO recycling and the adoption of alternative materials such as alloy-based electrodes, conducting polymers and ITO-free substrates, representing an active research direction[28–30,33]. Therefore, reducing the cost of PAL materials is a critical pathway to improving the overall cost-effectiveness of OPVs[13].

Organic semiconductor materials are typically synthesized through multiple steps that involve various catalytic cross-coupling reactions[34–36]. Of particular significance is the direct utilization of aryl C–H bonds as coupling partners, a strategy that has substantially reduced materials costs over the past few decades. This approach, commonly referred to as direct heteroarylation polymerization in polymer synthesis, not only reduces toxic by-products (for example, organotin compounds), but also eliminates two synthetic steps compared with traditional Stille or Suzuki–Miyaura protocols, avoiding the use of hazardous reagents including trimethyltin chloride and *n*-BuLi (ref. 34). It is worth noting that polythiophene derivatives synthesized by direct heteroarylation polymerization have recently achieved PCEs exceeding 16% (ref. 36). Further extension of this C–H arylation methodology to state-of-the-art OPV materials offers a feasible route towards cost-effective and sustainable materials synthesis[27].

Beyond the cost-reduction strategies enabled by materials synthesis methodologies, the synthetic complexity index has been applied to organic semiconductor materials and used as a cost metric to evaluate the trade-offs among cost, efficiency and stability in OPVs (synthetic complexity calculations are detailed in Supplementary Tables 1 and 2)[37]. High-performing OPVs have so far relied on polymer donor and NFA systems[10]. Figure 1c,d summarizes the development of key PAL materials with their synthetic complexity values, alongside the PCEs achieved in devices. For polymer donors, while achieving high PCEs typically necessitates complex synthesis steps (high synthetic complexity values), notable progress has been made in alleviating this interdependence[13], as demonstrated by low-cost materials such as PTQ10 (ref. 38), PTVT-T (ref. 39) and poly(thiophene) derivatives with synthetic complexity values below 35% (ref. 36). For acceptor materials, the introduction of dithienothiophen[3,2-*b*]pyrrolobenzothiadiazole (BTP)-based NFAs, such as Y6 (refs. 40,41), L8-BO (refs. 4,42), L8-BO-C4 (ref. 3) and AT-β2O (ref. 2), has enabled OPV PCEs to surpass 15%, and now even 20% (refs. 2,3,6). However, achieving PCEs beyond 19% continues to require acceptor materials with high synthetic complexity values (for example, 62.87% for Y6 and 97.50% for L8-BO-C4)[3,43], underscoring the challenge of balancing efficiency with synthetic simplicity. The incorporation of non-fused rings is crucial for both acceptor and donor materials (such as TBT-26 and PTQ18) to achieve lower synthetic complexity values (Fig. 1c,d), because the synthesis of fused rings typically involves more complex and costly procedures. Therefore, the combination of fully non-fused-ring acceptors and donors in PALs offers a viable way to reduce materials costs[44], although research on such high-performance PAL systems (PCE > 20%) is limited.

It should be noted that the contributors to materials costs include the price of starting or raw materials, synthetic complexity and energy

Fig. 1 | Cost analysis of OPV manufacturing. a, Manufacturing costs of OPV modules fabricated via high-speed and layer-by-layer (LbL) sequential deposition. The modules feature a device area of 9.0 cm^2, with the PAL covering 7.5 cm^2. Details of the low-cost, high-speed deposition manufacturing approach can be found in ref. 31. BHJ, bulk heterojunction. **b**, Materials costs of OPV modules. Left: proportion of material costs in the overall LbL module production process. Right: proportion of PAL material costs within total material costs. ETLs and HTLs refer to electron transport layers and hole transport layers, respectively. **c**, Progress in polymer donor materials, showing synthetic complexity alongside achieved PCEs. **d**, Progress in acceptor materials, illustrating synthetic complexity and the corresponding PCEs. The *x* axes in **c** and **d** denote the year each material was reported in OPV applications. The plotted PCE values represent the efficiencies achieved using these materials in OPV devices. The alkyl chains are identified as follows: R_1 = –CH_2–CH–(C_2H_5)(C_4H_9); R_2 = –CH_2–CH–(C_6H_{13})(C_4H_9); R_3 = –CH_2–CH–(C_6H_{13})(C_8H_{17}). The full names and chemical structures of the materials are summarized in Supplementary Note 1 and Supplementary Fig. 1, respectively. Further details of the synthetic complexity and PCE data are provided in Supplementary Figs. 2–18, Supplementary Tables 1 and 2, and Supplementary Note 2. **e**, PCE variations versus polymer molecular weight (M_w) for two PM6 polymers from different sources, D18, PM7, PBDB-T4Cl5S, PBD-Cl, PTQ10, DP1, DP3, PBCT-2F, SZ5, PM1 and D18-Cl (left to right; the *x*-axis label and values for PM6 apply to all 13 plots) and polydispersity (PDI) for PM6. The vertical and horizontal error bars represent the range from minimum to maximum values. Further details of the M_w, PDI and PCE data are provided in Supplementary Table 3. The PTQ10-based blends were blade-coated based on the inverted device architecture[45]. **f**, Approaches for reducing the costs of polymer donor materials. RT polymerization refers to synthetic polymerization methods conducted at ambient temperature (for example, room-temperature Suzuki–Miyaura polymerization)[34,35]. DHAP, direct heteroarylation polymerization.

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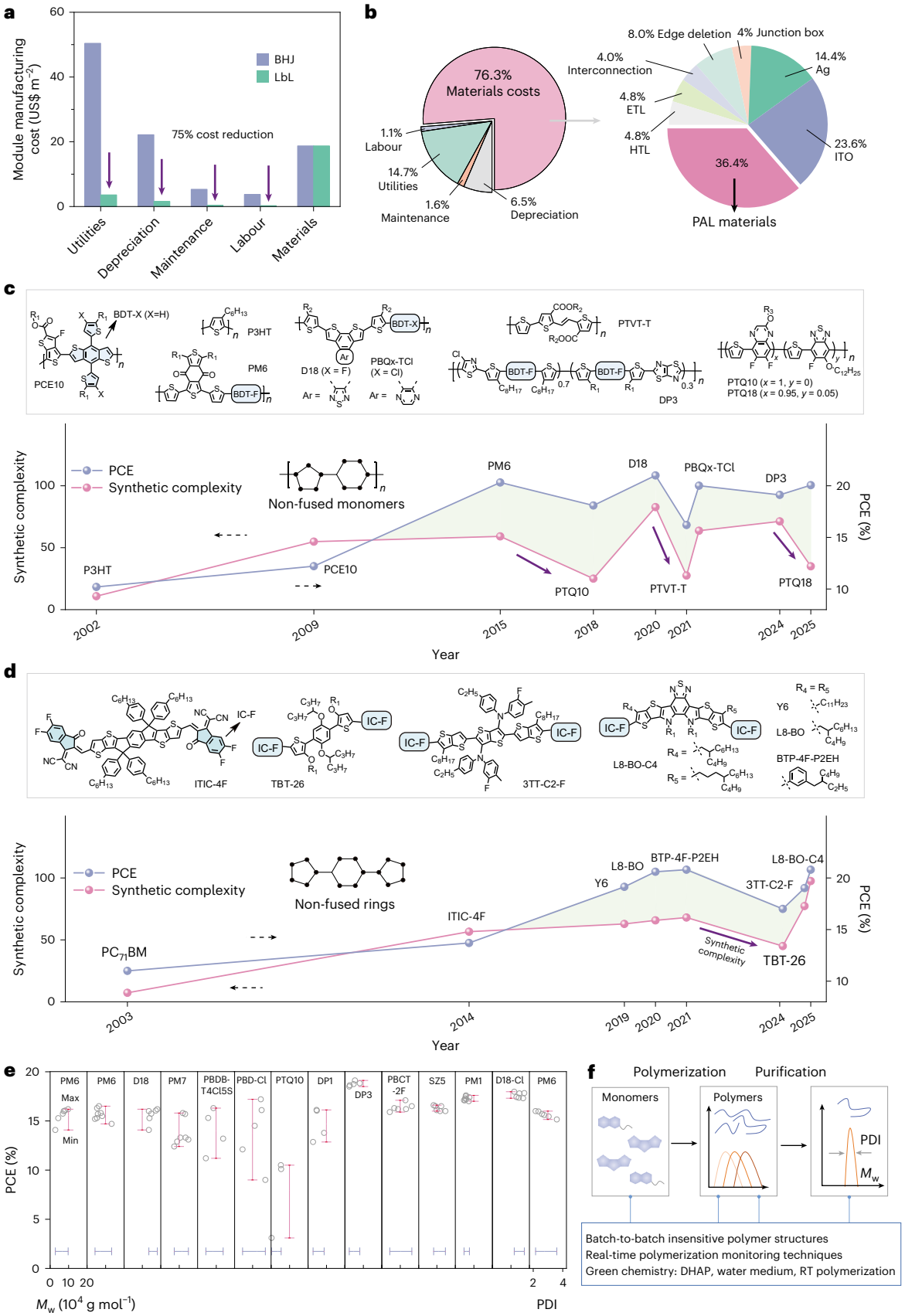
consumption during materials synthesis and production. While the synthetic complexity index provides a useful metric for assessing synthetic difficulty, it does not account for the raw materials costs and energy consumption. We also gathered pricing information for these PAL materials from commercial suppliers (detailed in Supplementary Table 1); the prices for donor and acceptor materials have increased since the fullerene age, highlighting the need to develop cost-effective alternatives. Polymer donors in particular



have low synthetic complexity values but generally higher costs than small-molecule counterparts owing to additional purification steps and batch-to-batch variations. To reduce the cost of polymer materials, it is essential to develop batch-consistent polymer materials and to elucidate the structural factors that contribute to batch-to-batch insensitivity. Recent advances have yielded several batch-to-batch-insensitive polymer systems that exhibit PCE variations below 1%, such as PM1 and DP3 (Supplementary Table 3). However, this batch-to-batch insensitivity has so far been demonstrated in only a limited number of polymer materials, and extending it to a wider range of OPV polymers is an ongoing challenge. Even with identical polymer structures, considerable PCE variations can occur owing to differences in M_w and polydispersity values[45,46], with PCE values ranging from 14.08% to 16.19% for PM6, from 3.1% to 10.5% for PTQ10 and from 12.86% to 16.12% for DP1 (Fig. 1e). The implementation of real-time polymerization monitoring techniques can address irreproducibility issues in high-performing OPVs (Fig. 1f)[47]. It is well established that the cost of organic materials decreases with industrial scale-up, as shown by experience in the pharmaceutical industry. With increased industrial effort, it is anticipated that risks associated with irreproducibility can be minimized through the development of advanced production and monitoring methodologies.

Recent studies have highlighted the prospects of high-performance OPV materials for cost reductions[13], particularly through strategies such as energy-efficient synthetic methods (such as room-temperature Suzuki–Miyaura polymerization)[35], the design of high-efficiency materials with simple structures[39,43] and the use of materials with high recyclability[33]. Other types of PAL material are also being explored and show encouraging efficiency and stability, such as oligomers and single-component materials[10]. Meanwhile, although the materials costs of transport layers account for 9.8% of the total (Fig. 1b), recent studies suggest that reducing the costs of transport layer materials and enhancing their large-scale processability can also benefit OPV manufacturing[48]. As PCEs now approach 21% (refs. 2,5,6,49), reducing materials costs without compromising PCEs is an important approach for enhancing the economic feasibility of OPVs. These high-performing materials should feature simple building blocks with minimal substitution sites (for example, polythiophene-based low-cost donor materials and non-fused NFAs) and be synthesized through environmentally benign, economical pathways[13,27,34,35]. It is important to recognize that the overall cost of OPVs includes not only materials costs but also expenses related to production plants, scale economies and other factors[32]. For example, the absence of aluminium frames, which are commonly used in silicon photovoltaic installations, can partially offset the materials costs of OPVs. Scale-up cost analyses, such as the levelized cost of electricity model[32], indicate that the cost of OPVs can be reduced with decreases in materials costs and the maturation of production technologies.

Materials processing with green solvents

After synthesis, the next step is processing OPV materials for their integration into devices. Conventional materials processing into thin films relies on chlorinated and aromatic solvents, which pose environmental and health risks as a consequence of their toxicity, volatility and waste management challenges. As illustrated in Fig. 2, solvents for processing organic materials can be classified into four groups[50]. The toxicity of these solvents to both the environment and human health can be assessed using the median lethal dose (LD50), lethal concentration 50% (LC50), half-maximal effective concentration (EC50), occupational exposure limit (AGW) and saturated vapour pressure (SVP) values. We plotted the LD50, LC50, EC50, AGW and SVP values alongside the PCEs achieved in single-solvent-processed OPV systems (Fig. 2b). Generally, solvents without halogen atoms and aromatic structures are considered greener. High LD50, LC50, EC50 and AGW values indicate low toxicity and higher safety for both the environment and human health. Group 2 solvents (non-halogenated) exhibit higher LD50 values than group 1 solvents (halogenated). However, it is crucial to note that structures without halogenated atoms and aromatic rings do not always indicate greener solvents, as some solvents can cause irreversible damage to human health. For instance, CS_2 and *N*-methyl-2-pyrrolidone, categorized under group 3 (non-halogenated and non-aromatic solvents), exhibit low LC50, EC50 and AGW values, highlighting their danger to the human nervous and reproductive systems. Even some alcohol-type solvents, such as 1-hexanol (group 4), show relatively high toxicity based on their LD50, LC50 and EC50 values. Therefore, solvents identified as green solvents for processing organic semiconductor materials may vary across the literature.

In this Review, we emphasize the use of bio-renewable solvents, such as 2-methyltetrahydrofuran (2-MeTHF)[51], eucalyptol[8,52], cyclopentyl methyl ether (CPME) and ethanol. These solvents can be derived from bio-resources and used for OPV manufacturing, efficiently reducing carbon footprints (Fig. 2a)[8,51,52]. Moreover, bio-renewable solvents generally exhibit high LD50, LC50, EC50 and AGW values. Their low SVP values indicate favourable safety profiles for practical applications. These solvents have been identified as greener alternative products available from commercial suppliers, such as Sigma-Aldrich.

By modifying the chemical structures of OPV materials to improve their compatibility with green solvents, high-performing OPV devices have achieved PCEs exceeding 19% using solvents such as tetrahydrofuran (THF), *o*-xylene and toluene[53]. Furthermore, mini-emulsification of conjugated materials in aqueous media (water mixed with organic solvents) has also led to PCEs over 20% (ref. 54), representing an encouraging advance in OPV materials with water processability. However, the reported PCEs for fully bio-renewable green-solvent-processed systems remain below 16% (Supplementary Table 4)[51,52]. The Hansen solubility parameters (HSP) method serves as a powerful support tool for predicting materials solubility and identifying alternative solvents, and has proved to be effective in developing bio-renewable solvent-processed PAL materials[8]. However, the HSP method has inherent limitations, as it relies on approximations obtained from experimental fitting or simulations based on group contributions. This HSP limitation is particularly relevant to polymers with varying M_w and crystallinity, which restricts its broader applicability. Side-group engineering through the incorporation of polar groups in PAL materials, along with the strategic use of guest molecules or additives, offers emerging avenues to enhance bio-renewable solvent compatibility while promoting environmentally sustainable processing[2,53,55]. The incorporation of polar side chains (for example, hydrophilic oligo(ethylene glycol) chains) reflects a shift away from the conventional non-polar side-chain strategies that have dominated conjugated OPV materials design for the past two decades. Establishing a standard for calculating the HSP of new materials, along with a unified figure of merit that considers PCEs, solvent toxicity and environmental impact, could facilitate better comparisons of the overall performance of state-of-the-art OPV materials systems. Given the extensive parameter space and the complexity of materials optimization required at this stage, integrating machine learning and high-throughput screening methodologies represents a promising direction to accelerate materials discovery[56,57].

Materials design with device longevity

Following the synthesis and processing of OPV materials into device architectures, the performance and stability of the resulting devices become important factors in their practical implementation. The performance of OPVs can degrade under operational conditions, closely linked to the stability of electrodes, interfaces and PALs[58]. The organic conjugated materials in PALs are formed through both covalent bonding and intermolecular interactions. Covalent bonds define the chemical structures of conjugated materials, enabling π-electron delocalization along the backbone and promoting charge transport. Simultaneously, intermolecular interactions, such as van der Waals forces, π–π stacking and dipole–dipole interactions, play a vital role in molecular self-assembly, ordering and the formation of materials[59]. During device operation, OPV materials encounter various stability

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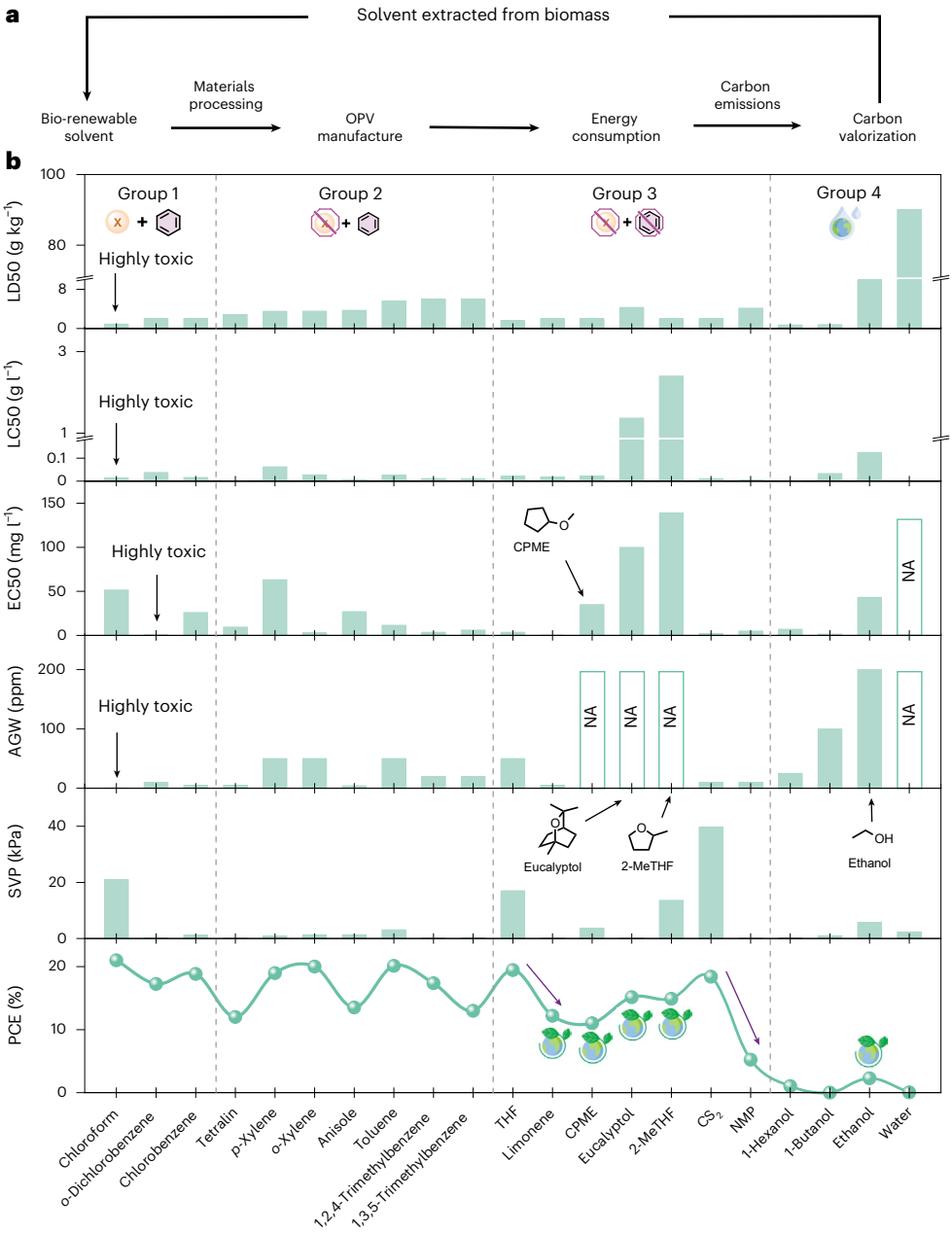

Fig. 2 | Green-solvent-processed OPV materials. a, Circular carbon cycle for organic materials processed with bio-renewable and greener alternative solvents. **b**, Classification of solvents for processing OPV materials and the PCEs achieved in devices. NMP, *N*-methyl-2-pyrrolidone. Halogenated solvents (group 1) typically dissolve donor and acceptor materials effectively, facilitating optimal phase separation and film formation. In contrast, many non-halogenated and non-aromatic solvents (group 3) exhibit poor solubility and unfavourable evaporation dynamics, often compromising film quality and device efficiency. The LD50 (for rats via oral intake) and LC50 (for rats via inhalation, vapour) values were sourced from the material safety data sheets from Sigma-Aldrich (www.sigmaaldrich.com; see Supplementary Table 4 for materials and catalogue numbers). The EC50 (for *Daphnia magna*), AGW (Arbeitsplatzgrenzwerte in Germany) and SVP values were collected from the material safety data sheets from Thermo Fisher Scientific (www.thermofisher.com) and Sigma-Aldrich (www.sigmaaldrich.com; see Supplementary Table 5 for materials and catalogue numbers). The chemical structures of the bio-renewable solvents are highlighted accordingly, and these bio-renewable solvents have also been identified as greener alternative products by commercial suppliers, such as Sigma-Aldrich. The chemical structures of the solvents are presented in Supplementary Fig. 19. NA indicates that the values are not applicable due to the intrinsically low toxicity and environmental friendliness. The purple arrows indicate decreases in PCEs and the incompatibility of PAL materials with bio-renewable solvents, bio-renewable and greener alternative solvents are marked by symbols.

challenges, including thermal, photo-and mechanical stability, as well as stability under ambient air conditions[60], which are intrinsically linked to these interactions and discussed below.

Materials degradation under thermal stress

Owing to the relatively weak intermolecular interactions and high mobility of soft side chains in OPV semiconductors, these molecular architectures tend to transition towards their lowest energy conformations under thermal stress, leading to morphological instability in as-deposited materials[59]. According to the International Summit on Organic Photovoltaic Stability protocol, the highest temperature for OPV stability testing is 85 °C (ref. 61), and the operating temperature of OPV devices is generally below the thermal decomposition threshold of organic semiconductor materials[62]. Consequently, the primary

concern for OPVs is thermally induced morphological instability, where disrupted molecular packing and phase separation in semiconductor blends degrade device performance[59,63]. It is worth noting that OPV devices may still experience high-temperature stress under specific conditions, particularly during high-temperature lamination encapsulation processes (for example, at 140–170 °C)[64]. For some applications, such as those in space, OPV materials that are resistant to high temperatures (>150 °C) are also required[22,65]. Therefore, in addition to the standard stability testing at 85 °C, researchers should carefully consider the extreme operational and processing conditions that the devices may encounter during both manufacturing processes (such as encapsulation) and end-use applications.

During the fullerene era, OPV devices were fabricated using fullerene derivatives as acceptors, such as [6,6]-phenyl-C_{61}-butyric acid methyl ester ($PC_{61}BM$) and [6,6]-phenyl-C_{71}-butyric acid methyl ester ($PC_{71}BM$). As a consequence of the low thermal diffusion energy of fullerenes, the migration of fullerene molecules within the PALs can lead to performance degradation in devices under thermal stress. A similar thermal diffusion effect has been observed in the widely studied conjugated polymer donor poly(3-hexylthiophene) (P3HT)[63]. The high content of isolated carbon–carbon single bonds (also lacking fused rings) along the conjugated polymer backbones and the presence of flexible side chains results in a low glass transition temperature (T_g) with reduced thermal diffusion energy.

With the transition to the non-fullerene era, acceptor–donor–acceptor (A–D–A)- and A–D–A–D–A-type multiple fused-ring acceptors with strong intermolecular interactions have been developed[66–68]. These acceptors exhibited improved thermal stability, effectively suppressing the thermal diffusion effect in OPV blends compared with fullerene derivatives[66]. Similar improvements have been observed in polymer donors such as PM6 and D18, in which stiff and rigid polymer backbones (high T_g) and strong intermolecular interactions restrict molecular diffusion to a greater extent than in P3HT (ref. 59). As a result, the thermal stability of OPV devices has been enhanced, as reported in the literature[68]. However, it should be noted that the presence of side chains in conjugated materials can induce thermal vibrations and, together with the diffusion-limited crystallization of non-fullerene acceptors below T_g, contribute to device degradation over time[66].

Given the multicomponent nature of PALs, their morphological stability is also governed by the Flory–Huggins interaction parameter (χ, which determines miscibility) and by molecular diffusion coefficients that control component mobility[63]. Thermodynamic stabilization occurs in low-χ systems (with high miscibility) through near-equilibrium mixing, whereas high-χ systems (with low miscibility) achieve kinetic stabilization through molecular immobilization. Optimal stability emerges when suppressed diffusion combines with χ values near the percolation threshold, effectively inhibiting phase segregation during device operation. Such thermodynamic behaviour can also be inferred from the cold crystallization peaks (T_{cc}) associated with NFA crystallization in PALs, reflecting its correlation with optimal device stability[67,69]. Overall, although the temperature-dependent χ shows a correlation with diffusion activation energies and helps predict OPV stability[63], there remains a lack of quantitative and reliable methods for describing the above molecular interaction effects in PALs[59,70], despite the critical need.

Progress has been made in enhancing the intrinsic thermal stability of OPV materials. However, the long-term stability of devices under continuous thermal stress remains a concern[59,62]. The stability of electrodes and interfaces constitutes another critical factor, including the intrinsic thermal diffusion of MoO_x and electrodes and the instability of poly(3,4-ethylenedioxythiophene):polystyrene sulfonate (PEDOT:PSS) layers (for example, the indium from ITO and Al can diffuse into PEDOT:PSS layers)[60]. Furthermore, studies have shown that, as the PAL materials interact with various organic/inorganic interlayers or electrodes, thermal diffusion across interfaces at PAL boundaries is an important issue that compromises device stability during long-term device operation (see details in Supplementary Fig. 20)[71]. Improving the crystallinity of PAL materials or cross-linked PAL materials has been shown to suppress thermal diffusion at interfaces[60]. Given that photoactive materials do not function as an isolated layer within a device, and despite mitigation strategies for PAL materials, developing immobilized interlayers or electrodes or using blocking buffer layers are also essential for improving the long-term stability of OPV devices[71–73].

Materials degradation under illumination

Ensuring the photostability of OPV materials is a fundamental requirement for the commercialization of OPV technology. Unlike thermal stability, photostability is governed by the photophysical and photochemical processes intrinsic to OPV materials. These processes include photoisomerization, covalent bond scission and the formation of new chemical bonds and free radicals, all of which could contribute to device degradation[60,74,75].

In the presence of environmental species such as O_2 and H_2O, photostability becomes even more complex. The photo-oxidation process and interactions with singlet oxygen ($^1O_2$) can accelerate the photodegradation of materials, thereby leading to performance losses in devices. This issue is particularly pronounced in polymer donors[76–78], where electron-rich units are highly susceptible to oxidation under illumination. Trapped $^1O_2$ within the OPV blends can also impact photostability by interfering with the photophysical processes in PALs[77]. Therefore, the pursuit of OPV materials with long-term photostability in air has been a persistent challenge over the past decades[78]. Even in the absence of O_2 and H_2O, the intrinsic photostability of OPV materials is still controlled by multiple photochemical pathways, including *cis–trans* photoisomerization of C=C bonds[74], bond scissions and hydrogen abstraction in side chains[21,75,79]. Moreover, given that OPV materials are in contact with various organic or inorganic interlayers (for example, ZnO) in which photocatalytic effects and interfacial photochemical reactions can occur, suppressing the existing thermal and photodegradation of OPV materials at different interfaces is essential to realize OPV devices with long-term stability[60,71,73].

With the help of robust encapsulation techniques, OPV devices have shown excellent operational stability in the absence of environmental species, leading to substantially extended lifetimes. T_{80} device lifetimes (defined as the time required for the PCE to decline to 80% of its initial value) of up to 30 years (and longer) under continuous illumination have been reported[73,80], and outdoor operational stability of 7 months for NFA-based OPV devices has been documented[79,81]. As the stability of OPV devices improves, tracking materials and device stability over extended periods (or even over tens of years) becomes more challenging. Fitted T_{80} lifetimes, typically determined using simple linear or exponential decay models, also face challenges in accurately predicting the long-term stability of OPV devices (Fig. 3a). Complex degradation mechanisms can lead to deviations from the trends extrapolated from limited stability data points. This highlights the need to acquire real-world operational data and to develop accelerated ageing methods that are applicable to both outdoor and indoor testing. Figure 3b,c illustrates the accelerated ageing method. The acceleration factor (ξ_{AF}) is extracted from degradation curves obtained under increased stress during a short test period (T_{test}). Fitting ξ_{AF} enables the extrapolation of long-term T_{80} lifetimes, allowing the prediction of device stability from short-term measurements[73,80,82] (Supplementary Fig. 21). Standardization in these areas will enable the establishment of robust testing protocols for assessing device failures under actual operating conditions (Fig. 3c), thereby enhancing understanding of OPV durability through the correlation of stability data with degradation mechanisms identified through accelerated ageing testing[82].

Materials degradation under mechanical stress

Owing to the unique covalent bonding and intermolecular interactions inherent in organic semiconductor materials, compared with

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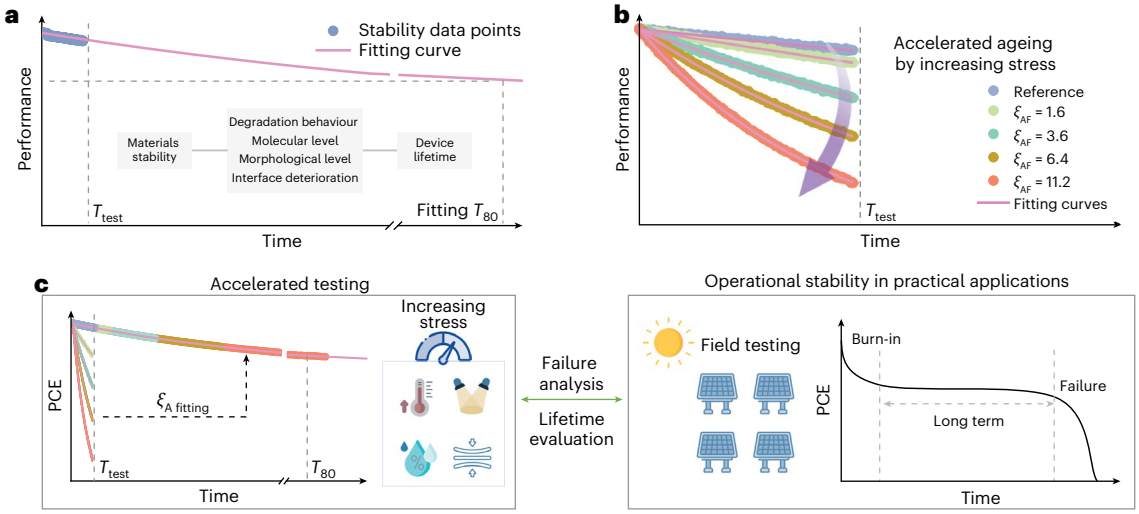

Fig. 3 | OPV materials stability and device lifetime. a, Device lifetime T_{80} prediction using linear or exponential decay models. The intrinsic stability of materials at both the molecular and morphological levels affects device lifetime, alongside stability challenges of OPV materials in interactions with interfaces and electrodes (photochemical and thermal degradation effects; see Supplementary Fig. 20 for more details). **b**, Accelerated ageing method for predicting long-term stability. Derived from the degradation curves, ξ_{AF} quantifies the rate at which the degradation process is accelerated under elevated stress conditions compared with normal operating conditions. This factor allows the prediction of the device lifetime by scaling the accelerated test results to real-world scenarios. The data points were computer-generated by adding zero-mean Gaussian noise with a standard deviation of 0.1 to the single-exponential decay functions represented by the fitting curves. **c**, Accelerated ageing method (left) and operational stability under practical conditions (right). By applying increased stress during a short testing period (T_{test}), long-term T_{80} lifetimes can be extrapolated from the fitted degradation curves, enabling the prediction of long-term stability from short-term tests (Supplementary Fig. 21).

inorganic perovskites and silicon, OPV devices offer the advantage of mechanical flexibility. This mechanical flexibility has been demonstrated in applications such as flexible sensors, wearable electronics and lightweight photovoltaic devices[7,12,19,83,84]. Recent studies have reported that OPV device performance can even be enhanced under mechanical stress[85]. While perovskite- and silicon-based photovoltaics have achieved notable progress in flexible electronics, organic polymeric materials exhibit exceptional promise for flexible electronic applications owing to their ability to achieve conformal integration with curved surfaces, stretchable substrates and wearable devices. Molecular engineering on polymer structures, including side-chain modification via hydrogen bonding or siloxane-terminated groups, along with backbone modulation through conjugation break spacers, can efficiently enhance the mechanical resilience of these materials while preserving optoelectronic performance[84]. Therefore, among the photovoltaic technologies, polymer material-based OPVs are the most viable and compelling option for flexible electronic applications[12,19,85,86].

Materials design with device efficiency

Efficiency is a critical parameter for OPVs, reflecting their ability to convert light into electrical energy. Recent advances in materials design, including the development of NFAs, donor materials, charge-selective layers and ternary blends, have led to improvements in device efficiency[5,10,42,48,87]. As shown in Fig. 4a, state-of-the-art OPV systems have achieved PCEs over 20% (refs. 2,5,6). Unlike inorganic semiconductors, organic semiconductors usually form coulombically bound excitons when photoexcited owing to the low dielectric constant of those materials[88,89] (Fig. 4b). Consequently, compared with single-junction perovskite and silicon solar cells, OPVs suffer more efficiency losses, primarily as a result of the relatively higher voltage loss with non-radiative charge recombination[88–90] (Supplementary Table 6). Although the low dielectric constants and large capture radii of organic materials (which differ from those of perovskite materials by an order of magnitude) pose challenges, approaches such as developing organic materials with intrinsically high luminescence and effectively re-dissociating spin-triplet exciton states into free charges have been shown to reduce non-radiative voltage losses[3,90,91]. Focusing on these strategies to reduce energy loss provides a viable avenue to further enhance PCEs beyond current limits.

State-of-the-art materials systems with PCEs over 20% typically use PALs that are 100–130 nm thick and have small device areas of 0.04–0.1 cm^2 (refs. 2,4–6). However, for practical applications, producing continuous and uniform large-area films of organic semiconductors at this thickness presents manufacturing challenges[92]. OPVs experience efficiency losses when transitioning from thin to thicker layers, primarily owing to the limited exciton diffusion lengths and increased charge recombination[2] (Fig. 4c and Supplementary Table 7). High-performance thick PALs place stringent demands on polymer donors, which intrinsically exhibit shorter exciton diffusion lengths (L_D) than NFAs. Therefore, developing high-performance polymer donors with both long L_D and well-balanced, high-charge-transport properties in PALs is essential for achieving thickness-insensitive OPV devices. It should be noted that the materials systems in Fig. 4c are both high-cost materials with multi-fused rings. The values of PCE/($SCP_{D:A}$ × thickness), where SCP represents the synthetic complexity of D:A materials and thickness refers to the PAL thickness, demonstrate that the cost-efficiency disparity in thick-film PALs is more pronounced than that of thin-film PALs[45,93] (Supplementary Tables 8 and 9). Similarly, scaling up from small-area to large-area devices also leads to PCE losses, largely driven by challenges in maintaining uniform morphological properties in the organic semiconductor layers during manufacturing (Fig. 4d and Supplementary Table 10).

Efforts to scale up OPV devices in laboratories have occasionally relied on interlayers that are incompatible with industry standards. This lab-to-fab gap arises not only from challenges related to PAL materials, but also from interface and electrode compatibility issues that limit the transfer of high PCEs from small-area lab devices to thick, large-area modules. Therefore, we argue that instead of intensifying research pushing PCEs in small-area PAL systems, close collaboration between academia and industry is essential to enhance the scalability and cost-effectiveness of high-performance materials systems for scalable manufacturing to realize the commercial viability of OPVs[1,92,94].

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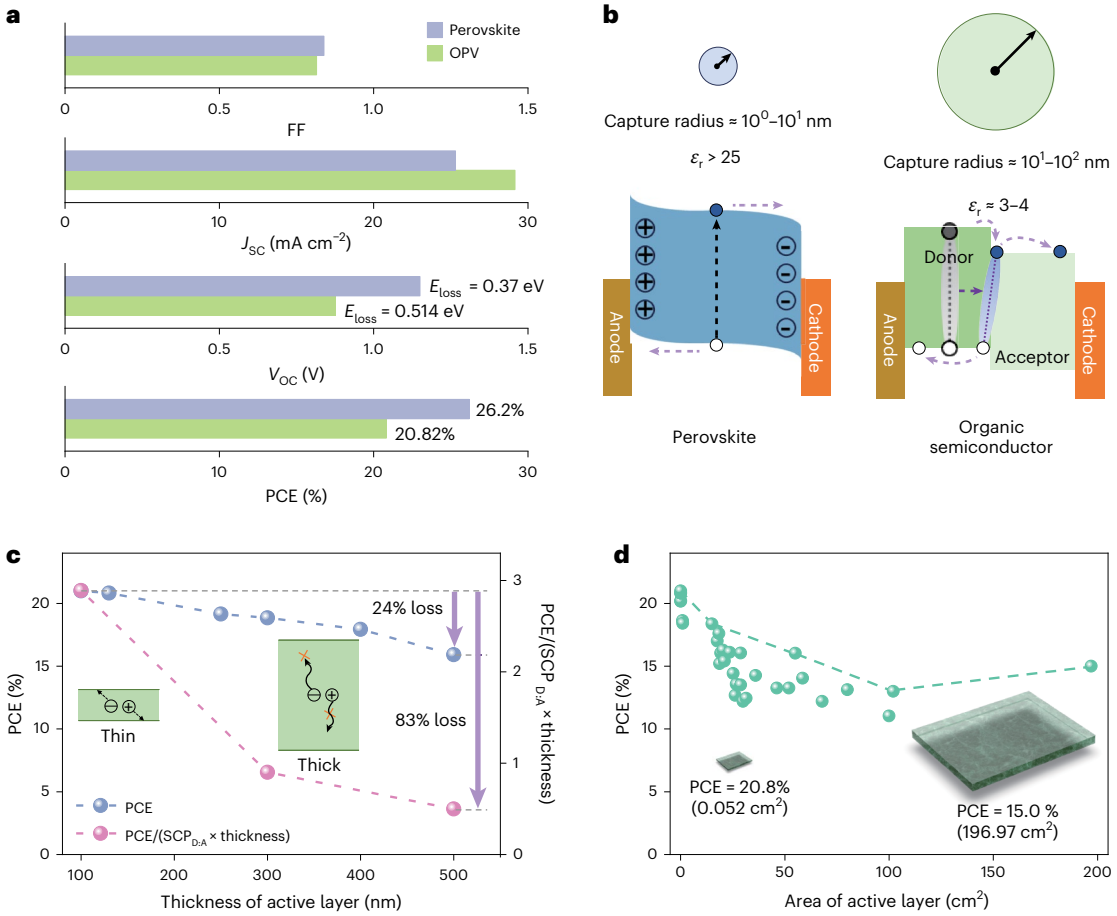

Fig. 4 | Challenges in device efficiency for OPV materials. a, Efficiency gap between OPV and perovskite materials in devices, highlighting the presence of intrinsic energy losses in OPV materials. FF is the fill factor, J_{SC} is the short-circuit current density, V_{OC} is the open-circuit voltage and E_{loss} is the energy loss. Further details of the materials systems and device parameters are available in Supplementary Table 6. **b**, The main photophysical process in perovskite and OPV materials. Perovskite materials generally exhibit a high dielectric constant (ε_r > 25) with a small Coulomb capture radius of 10^0–10^1 nm. However, OPV materials have a low ε_r ≈ 3–4 with a large Coulomb capture radius of 10^1–10^2 nm (ref. 89). **c**,**d**, PCEs achieved in state-of-the-art OPV materials with different PAL thicknesses (**c**) and different PAL areas (**d**). The parameter PCE/(SCP$_{D:A}$ × thickness), which is a metric that evaluates the efficiency of the PALs corresponding to each unit of synthetic complexity and thickness, thereby assessing performance across different materials and manufacturing conditions, is also shown in **c** (see Supplementary Tables 7–9 and Supplementary Figs. 22–25 for more details).

Materials adaptation for real-world applications

Leveraging the lightweight and mechanically flexible nature of organic semiconductors, OPV materials have achieved excellent stretchability, sustaining a strain of 95.5% in free-standing thin-films[19] and a power-to-weight ratio of up to 40 W g^{-1} (ref. 95). These advantages enable applications of OPV devices in portable and wearable electronics, such as thermoregulatory clothing, self-powered ultra-flexible wearable sensors and other self-powered electronics[7,12,83]. The easily tunable electronic properties of organic semiconductors allow OPVs to efficiently harvest indoor illumination energy at specific wavelengths, benefiting Internet of Things applications such as OPV-powered wireless devices for efficient power transfer[18,96,97]. Organic semiconductor materials can also be tailored to selectively capture solar energy at specific wavelengths, enabling semi-transparent OPVs to be applied in building-integrated photovoltaics and solar-integrated greenhouses for crop production[9,14]. Recent studies have also explored the applications of OPVs in underwater environments[25]. The lightweight and adaptable properties of OPV materials have attracted interest for applications in space and high-altitude platform stations[20,21,23] owing to their resilience in extreme environments. These PAL materials demonstrate reliable performance across a temperature range from −173 to 117 °C and robust radiation resistance[21,65].

It is important to recognize that different applications demand distinct properties, and optimizing one may come at the expense of another. For greenhouse and building-integrated photovoltaic applications, PAL materials should possess high average transmittance values in the visible range with low-bandgap characteristics. In contrast, indoor applications demand strong visible light absorption. For self-powered electronics in biological applications, thick high-performance PAL systems may not be necessary. Flexible and wearable applications may require soft conjugation breakers in polymer backbones to enhance the mechanical flexibility of PALs. However, these soft conjugation breakers can decrease the thermal stability of PAL materials, making them unsuitable for space applications. Therefore, to accelerate their adoption, materials design strategies must be tailored to these application-driven needs, and high-throughput screening—along with machine learning-assisted approaches—can facilitate optimal performance and integration with real-world applications[56,57,93].

Outlook

This Review examines the materials challenges that have hindered OPV commercialization. With PCEs now exceeding 21%, OPVs are closer than ever to commercialization. Bridging the cost–stability–efficiency gap and ensuring long-term sustainability are essential next steps. The research community has reached milestones with proof-of-concept demonstrations at a technology readiness level of 2 (TRL 2), showcasing

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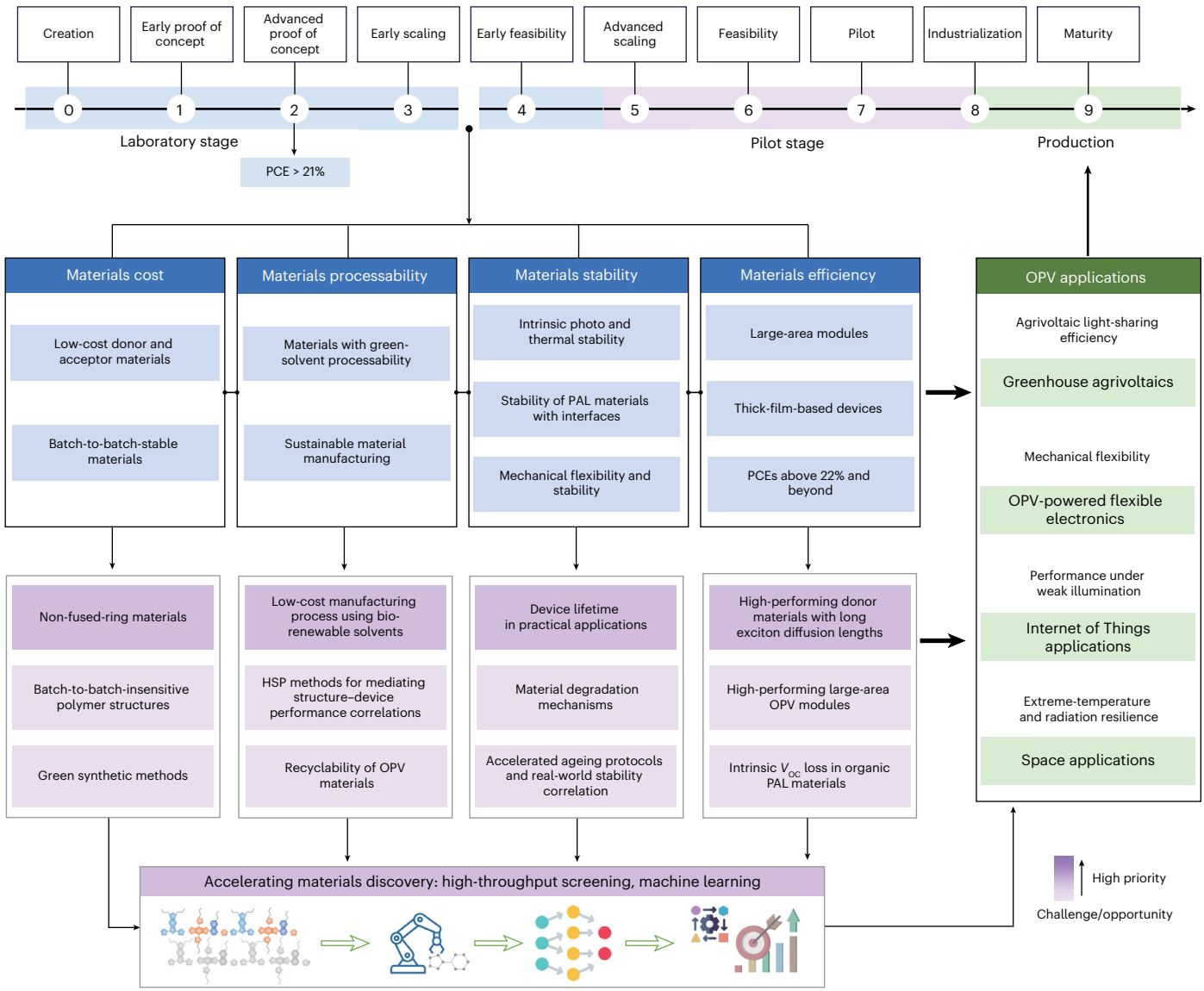

Fig. 5 | Commercial viability considerations for OPVs with fundamental challenges and potential opportunities in practical applications. Before commercialization, the technology must progress through successive TRLs, encompassing fundamental research and development at laboratory stage (TRLs 1–4), prototype validation and pilot-scale demonstration (TRLs 5–7) and ultimately full-scale production (TRLs 8 and 9). Dark purple shading highlights high-priority challenges and opportunities to accelerate the TRL progression. The development of OPV technologies has largely been confined to early TRLs (1–3) so far, primarily at the laboratory scale.

high PCEs and various prototype devices. However, advancing from laboratory-scale demonstrations (TRLs 0–4) to commercial deployment (TRLs 5–9) remains challenging, owing to the persistent trade-offs among cost, stability and efficiency when scaling up.

Building on the above analyses and discussions, we delineate the TRL progression of OPV developments (Fig. 5) and highlight several challenges that merit prioritization to facilitate commercialization and successful advancements across the TRL spectrum. The first challenge demands materials design strategies focused on low-cost, batch-to-batch-consistent materials and includes an emphasis on non-fused-ring PAL systems, such as high-performance polythiophene-based donor materials, which offer simpler synthesis and scalability advantages. Moreover, environmentally friendly synthetic routes, including direct heteroarylation polymerization, room-temperature processes and water-medium syntheses, are critical for reducing environmental impact and lowering production costs. The development of bio-renewable solvent-processable formulations and scalable manufacturing techniques is equally important. For example, solvents such as 2-MeTHF and eucalyptol have been demonstrated to be eco-friendly alternatives that minimize carbon footprints in large-scale manufacturing.

The second challenge involves elucidating the intrinsic degradation mechanisms of PAL materials under various stress factors (such as light, heat, oxygen exposure and electrical bias). Moreover, it is crucial to understand the stability of PAL materials in contact with interfaces and electrodes within devices, as interfacial interactions affect device longevity. Together, these insights will enable reliable assessments of the actual operational lifetime of OPVs under practical conditions. A notable knowledge gap remains: the relationship between accelerated ageing tests conducted in laboratories and the real-world performance of OPV modules is not well established. Bridging this gap is essential for accurate lifetime predictions, which in turn are vital for the deployment and commercialization of OPVs.

Despite these challenges, OPV materials offer advantages such as mechanical flexibility, sensitivity under low-light conditions and tunable transparency that make them suitable for diverse applications.

Realizing these advantages requires collaboration between academic researchers and industry, guiding academic research directions to address industrial challenges while simultaneously translating laboratory innovations into commercially viable technologies.

Finally, the extensive materials library accumulated over three decades of OPV research provides a foundation for future progress. To accelerate materials discovery, high-throughput screening and machine learning-assisted molecular design represent transformative strategies for optimizing next-generation materials[56,57,93]. Researchers must prioritize overcoming the cost–stability–efficiency trade-off during scale-up. Success hinges on advances in all of these areas to transition OPVs from laboratory prototypes into practical renewable energy solutions. Such collective and concerted efforts are crucial to unlocking the commercial potential of OPVs.

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Author contributions

D.B. and J.H. contributed to the conception, writing and editing of every section of the Review. J.H. drafted the initial paper, and H.X., D.C. and A.S. thoroughly edited and refined each section, providing valuable insights. All authors contributed to discussing, editing and reviewing the paper before submission.

Competing interests

The authors declare no competing interests.

Additional information

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